

TERM PROJECT

Final Report

GROUP - 9

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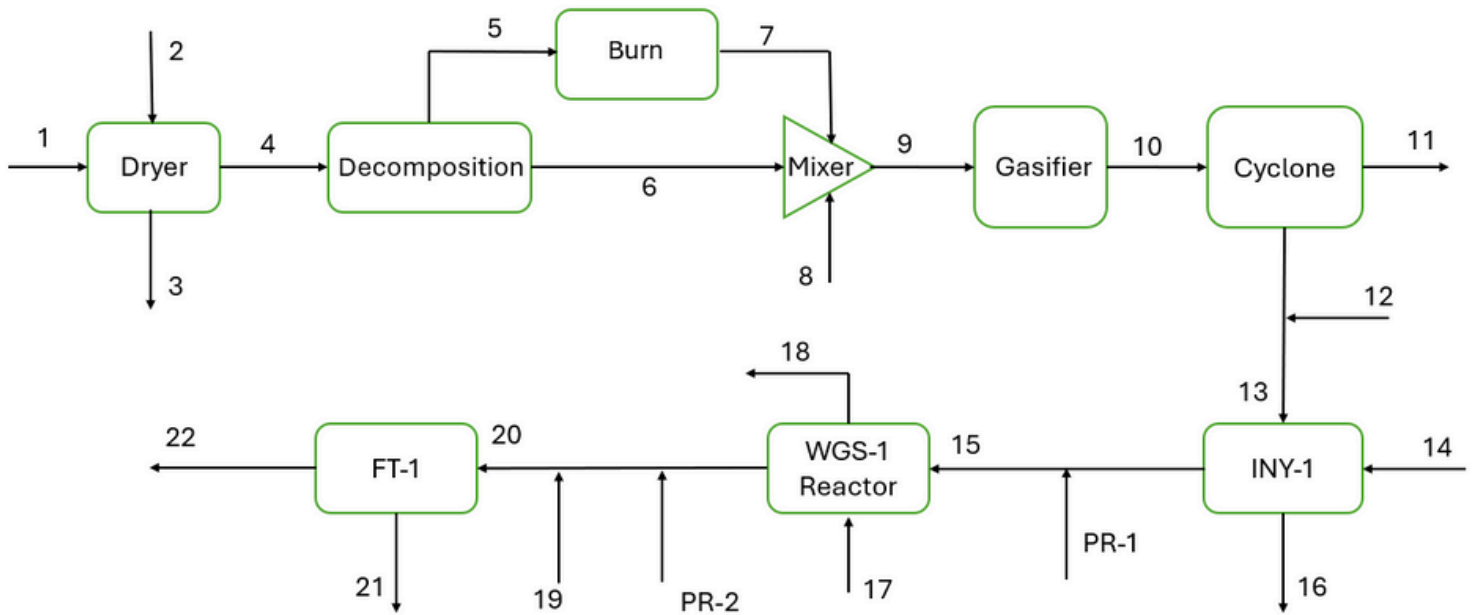
1. Background and Motivation

Hydrogen is emerging as a crucial clean energy carrier for achieving carbon neutrality. Biogas, primarily composed of methane and carbon dioxide, represents an abundant renewable resource that can be converted to high-purity hydrogen through steam reforming processes. This project focuses on developing a comprehensive process design for hydrogen production from biogas using steam reforming followed by water-gas shift reaction and pressure swing adsorption for purification.

2. Problem Statement

Producing high-purity hydrogen from biogas poses key challenges such as removing corrosive H_2S impurities, achieving **99.99%** fuel-cell grade purity, and optimizing operating parameters like steam-to-carbon ratio, temperature, and pressure for maximum yield and efficiency. This project aims to design and simulate a complete process flowsheet that integrates reforming, water-gas shift, CO_2 capture, and purification steps. The expected outcomes include an optimized flowsheet with mass balances, demonstration of hydrogen purity, and a comparison of performance and efficiency with conventional steam methane reforming (SMR).

3. Process Flowchart



Stream composition and flowrates (including variables) :

STREAM	FLOWRATE(kg/h)	COMPOSITIONS(all W_i^j are in kg/h and % in wt%)
1 ₁	7402.13	79.74% BIOMASS(C-27.35%, O-59.30%, S-0.65%, N-0.11%, H-5.82%, OTHER INORGANIC MATTER - 6.77%) 20.26% H ₂ O
2		Heat exchanger
3	1500	H ₂ O 100%
4 ₂	5902.13	DRIED BIOMASS (C- W_C^4 , H- W_H^4 , N- W_N^4 , S- W_S^4 , O- W_O^4 , INORGANIC MATTER- W_I^4)
5 ₃	5502.63	100% BIOMASS WITHOUT INORGANIC MATTER (C- W_C^4 , H- W_H^4 , N- W_N^4 , S- W_S^4 , O- W_O^4)
6	399.5	100% INORGANIC MATTER(W_I^4)

1. The flowrate and composition of every element has been changed from mid-sem report
2. and 3. The flowrate has been changed.

7**	8100.5	$\text{CO}_2\text{-}W_{\text{CO}_2}^7, \text{SO}_2\text{-}W_{\text{SO}_2}^7, \text{NH}_3\text{-}W_{\text{NH}_3}^7, \text{H}_2\text{O}\text{-}W_{\text{H}_2\text{O}}^7,$ $\text{CO}\text{-}W_{\text{CO}}^7,$ $\text{H}_2\text{-}W_{\text{H}_2}^7, \text{CH}_4\text{-}W_{\text{CH}_4}^7, \text{H}_2\text{S} - W_{\text{H}_2\text{S}}^7$
8*	F₈	Carbon(char)- $W_{\text{C}}^8, \text{CO}_2\text{-}W_{\text{CO}_2}^8, \text{H}_2\text{O}\text{-}W_{\text{H}_2\text{O}}^8$
9	F₉	$\text{CO}_2\text{-}W_{\text{CO}_2}^9, \text{SO}_2\text{-}W_{\text{SO}_2}^7, \text{NH}_3\text{-}W_{\text{NH}_3}^7, \text{H}_2\text{O}\text{-}W_{\text{H}_2\text{O}}^9,$ $\text{CO}\text{-}W_{\text{CO}}^7, \text{C(char)}\text{-}W_{\text{C}}^8, \text{H}_2\text{-}W_{\text{H}_2}^7, \text{ASH}\text{-}W_{\text{I}}^4, \text{CH}_4\text{-}$ $W_{\text{CH}_4}^7, \text{H}_2\text{S} - W_{\text{H}_2\text{S}}^7$
10	F₁₀	$\text{SO}_2\text{-}W_{\text{SO}_2}^7, \text{NH}_3\text{-}W_{\text{NH}_3}^7, \text{CO}\text{-}W_{\text{CO}}^{10},$ $\text{H}_2\text{-}W_{\text{H}_2}^{10}, \text{ASH}\text{-}W_{\text{I}}^4, \text{H}_2\text{S} - W_{\text{H}_2\text{S}}^7, \text{CH}_4\text{-}W_{\text{CH}_4}^7$
11	399.5	100% ASH
12		Heat exchanger
13	F₁₃	$\text{SO}_2\text{-}W_{\text{SO}_2}^7, \text{NH}_3\text{-}W_{\text{NH}_3}^7, \text{CO}\text{-}W_{\text{CO}}^{10},$ $\text{H}_2\text{-}W_{\text{H}_2}^{10}, \text{H}_2\text{S} - W_{\text{H}_2\text{S}}^7, \text{CH}_4\text{-}W_{\text{CH}_4}^7$
14	46778	100% Methyldiethanolamine ($\text{C}_5\text{H}_{13}\text{NO}_2$)
15	F₁₅	$\text{CO}\text{-}W_{\text{CO}}^7, \text{H}_2\text{-}W_{\text{H}_2}^7$
16	F₁₆	$\text{SO}_2\text{-}W_{\text{SO}_2}^7, \text{NH}_3\text{-}W_{\text{NH}_3}^7, \text{C}_5\text{H}_{13}\text{NO}_2\text{-}46,778 \text{ kg/h}, \text{H}_2\text{S}$ $- W_{\text{H}_2\text{S}}^7, \text{CH}_4\text{-}W_{\text{CH}_4}^7$
17	10000	100% H ₂ O

*all the components input are in such an amount that all the reactants in the reactions of the gasifier can get fully consumed.

**Water of amount 2597.87 KG/H has been added to the fifth after the reactions.

18	F_{18}	100% H_2O
19		Heat exchanger
20	F_{20}	$CO-W_{CO}^{20}, CO_2-W_{CO_2}^{20}, H_2-W_{H_2}^{20}, CH_4-W_{CH_4}^{20}$
21	F_{21}	$CO-W_{CO}^{20}, CO_2-W_{CO_2}^{21}, CH_4-W_{CH_4}^{20}$
22	P	$H_2-W_{H_2}^{22}, CO_2-W_{CO_2}^{22}$

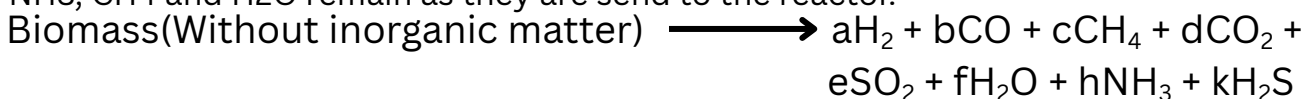
• Reasons for differences from Mid - sem report

1. As the composition of 7th stream was specified and the prior input was not giving the desired flowrate and composition (H to C ratio) in the 7th stream so it was changed.
2. and 3. As the flowrate of 1st stream changed so 4th and 5th streams' flowrate will also be changed.

Reactions Involved :

Burner(Pyrolysis)-:

33.1621% of C is converted to CO and 66.8379% in C in CO_2 and S is 100% converted. NH_3 , CH_4 and H_2O remain as they are send to the reactor.



Gasifier-:



- Single-pass conversion : 100%

($C + CO_2 \rightarrow 2CO$): ~40% carbon

($C + H_2O \rightarrow CO + H_2$): ~60% carbon

INY-1 :

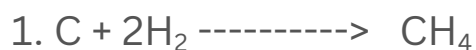


WGS-1 :



- Single pass conversion 90%

Decomposition:



single pass conv. 2.999559%

single pass conv. 100%

Additional Information :

1. Stream 7 composition

- CO₂: 47.5 wt%
- H₂O: 32.0 wt%
- CO: 15.0 wt%
- H₂: 4.0 wt%
- CH₄: 0.8 wt%
- H₂S: 0.4 wt%
- SO₂: 0.20 wt%
- NH₃: 0.10 wt%

2. Catalysts for reactions in gasifier

1. Iron(Fe)
2. iron oxide-chromium
oxide(Fe₂O₃-Cr₂O₃)

3. PR refers to pressure regulators

4. The Methyldiethanolamine used in stream 14 is used to remove components of stream 13 which are not CO and H₂

5. In FT-1 zeolites and silica gel are used as absorbents which absorb the impurities (CO,CO₂,CH₄) and releases them after the pressure is lowered and gets regenerated.

1.Assumptions :

1. Operation under steady-state.
2. Solid carbon is used as char.
3. Tar formation is not considered.
4. Biomass has entered as a nonconventional solid.
5. Ash has entered as a nonconventional solid.
6. Isothermal gasification.
7. Boudouard reaction consumes 40% carbon and 60% of carbon goes in water gas shift reaction in gasifier.

2.MATLAB

Streams :

2.1 1st stream weights

$$\begin{aligned}F_1 &= 7402.13 \text{ kg/hr} \\W_{biomass}^1 &= 0.7974 \times F_1 = 0.7974 \times 7402.13 = 5902.13 \text{ kg/hr} \\W_{H_2O}^1 &= 0.2026 \times F_1 = 0.2026 \times 7402.13 = 1500 \text{ kg/hr} \\W_C^1 &= 0.2735 \times W_{biomass}^1 = 0.2735 \times 5902.13 = 1614.02 \text{ kg/hr} \\W_O^1 &= 0.5930 \times W_{biomass}^1 = 0.5930 \times 5902.13 = 3499.92 \text{ kg/hr} \\W_S^1 &= 0.0065 \times W_{biomass}^1 = 0.0065 \times 5902.13 = 38.52 \text{ kg/hr} \\W_N^1 &= 0.0011 \times W_{biomass}^1 = 0.0011 \times 5902.13 = 6.66 \text{ kg/hr} \\W_H^1 &= 0.0582 \times W_{biomass}^1 = 0.0582 \times 5902.13 = 343.51 \text{ kg/hr} \\W_I^1 &= 0.0677 \times W_{biomass}^1 = 0.0677 \times 5902.13 = 399.5 \text{ kg/hr}\end{aligned}$$

2.2 4th stream weights

Note: Except water, all the components of Stream 1 are transferred to Stream 4.

$$\begin{aligned}W_i^4 &= W_i^1 = \text{value for all } i \in \{C, O, S, N, H, I\} \\W_C^4 &= W_C^1 = 1614.02 \text{ kg/hr} \\W_O^4 &= W_O^1 = 3499.92 \text{ kg/hr} \\W_S^4 &= W_S^1 = 38.52 \text{ kg/hr} \\W_N^4 &= W_N^1 = 6.66 \text{ kg/hr} \\W_H^4 &= W_H^1 = 343.51 \text{ kg/hr} \\W_I^4 &= W_I^1 = 399.5 \text{ kg/hr}\end{aligned}$$

2.3 5th stream weights

Note: All the components of Stream 5 transfer to Stream 4, except the inorganic matter (W_I).

There is no inorganic matter in this stream.

$$\begin{aligned}W_i^5 &= W_i^4 = \text{value for all } i \in \{C, O, S, N, H, H_2O\} \\W_C^5 &= W_C^4 = 1614.02 \text{ kg/hr} \\W_O^5 &= W_O^4 = 3499.92 \text{ kg/hr} \\W_S^5 &= W_S^4 = 38.52 \text{ kg/hr} \\W_N^5 &= W_N^4 = 6.66 \text{ kg/hr} \\W_H^5 &= W_H^4 = 343.51 \text{ kg/hr}\end{aligned}$$

2.4 7th stream weights

$$F_7 = 8100.5 \text{ kg/hr}$$

$$W_{CO_2}^7 = F_7 \times x_{CO_2}^7 = 8100.5 \times 0.475 = 3847.74 \text{ kg/hr}$$

$$W_{SO_2}^7 = F_7 \times x_{SO_2}^7 = 8100.5 \times 0.002 = 16.20 \text{ kg/hr}$$

$$W_{H_2}^7 = F_7 \times x_{H_2}^7 = 8100.5 \times 0.04 = 324.02 \text{ kg/hr}$$

$$W_{H_2O}^7 = F_7 \times x_{H_2O}^7 = 8100.5 \times 0.32 = 2592.16 \text{ kg/hr}$$

$$W_{CO}^7 = F_7 \times x_{CO}^7 = 8100.5 \times 0.15 = 1215.08 \text{ kg/hr}$$

$$W_{NH_3}^7 = F_7 \times x_{NH_3}^7 = 8100.5 \times 0.001 = 8.10 \text{ kg/hr}$$

$$W_{CH_4}^7 = F_7 \times x_{CH_4}^7 = 8100.5 \times 0.008 = 64.80 \text{ kg/hr}$$

$$W_{H_2S}^7 = F_7 \times x_{H_2S}^7 = 8100.5 \times 0.004 = 32.40 \text{ kg/hr}$$

2.5 8th stream weights

$$W_C^8 = 2880.177 \text{ kg/hr}$$

$$W_{CO_2}^8 = 376.524 \text{ kg/hr}$$

$$W_{H_2O}^8 = 0.00 \text{ kg/hr}$$

2.6 9th stream weights

Note: For Stream 9,

- $W_{CO_2}^9$ and $W_{H_2O}^9$ are the sum of corresponding weights from Streams 7 and 8.
- W_C^9 is equal to W_C^8 .
- All other components have the same weights as in Stream 7.

$$W_{CO_2}^9 = W_{CO_2}^7 + W_{CO_2}^8 = 3847.74 + 376.524 = 4224.26 \text{ kg/hr}$$

$$W_{H_2O}^9 = W_{H_2O}^7 + W_{H_2O}^8 = 2592.16 + 0 = 2592.16 \text{ kg/hr}$$

$$W_C^9 = W_C^8 = 2880.177 \text{ kg/hr}$$

$$W_{SO_2}^9 = W_{SO_2}^7 = 16.20 \text{ kg/hr}$$

$$W_{H_2}^9 = W_{H_2}^7 = 324.02 \text{ kg/hr}$$

$$W_{CO}^9 = W_{CO}^7 = 1215.08 \text{ kg/hr}$$

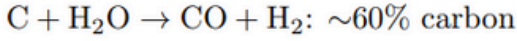
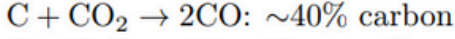
$$W_{NH_3}^9 = W_{NH_3}^7 = 8.10 \text{ kg/hr}$$

$$W_{CH_4}^9 = W_{CH_4}^7 = 64.80 \text{ kg/hr}$$

$$W_{H_2S}^9 = W_{H_2S}^7 = 32.40 \text{ kg/hr}$$

2.7 10th stream weights

Note (Gasifier): Single-pass conversion = 100%.



$$W_{CO}^{10} = W_{CO}^7 + \left[2(0.4) \left(\frac{W_C^9}{12} \right) + (0.6) \left(\frac{W_C^9}{12} \right) \right] \times 28 = 10623.66 \text{ kg/hr}$$

$$W_{H_2}^{10} = W_{H_2}^7 + (0.6) \left(\frac{W_C^9}{12} \right) \times 2 = 612.04 \text{ kg/hr}$$

$$W_{CO_2}^{10} = W_{CO_2}^9 - (0.4) \left(\frac{W_C^9}{12} \right) \times 44 = 0.00 \text{ kg/hr}$$

$$W_{H_2O}^{10} = W_{H_2O}^9 - (0.6) \left(\frac{W_C^9}{12} \right) \times 18 = 0.00 \text{ kg/hr}$$

$$W_C^{10} = 0.00 \text{ kg/hr}$$

$$W_{SO_2}^{10} = W_{SO_2}^9 = 16.20 \text{ kg/hr}$$

$$W_{NH_3}^{10} = W_{NH_3}^9 = 8.10 \text{ kg/hr}$$

$$W_I^{10} = W_I^9 = 399.50 \text{ kg/hr}$$

$$W_{CH_4}^{10} = W_{CH_4}^9 = 64.80 \text{ kg/hr}$$

$$W_{H_2S}^{10} = W_{H_2S}^9 = 32.40 \text{ kg/hr}$$

2.8 13th stream weights

Note: All the components of Stream 13 are identical to Stream 10, except for the ash component (W_{Ash}), which is removed in the cyclone.

$$W_{CO}^{13} = W_{CO}^{10} = 10623.66 \text{ kg/hr}$$

$$W_{H_2}^{13} = W_{H_2}^{10} = 612.04 \text{ kg/hr}$$

$$W_{CO_2}^{13} = W_{CO_2}^{10} = 0.00 \text{ kg/hr}$$

$$W_{H_2O}^{13} = W_{H_2O}^{10} = 0.00 \text{ kg/hr}$$

$$W_C^{13} = W_C^{10} = 0.00 \text{ kg/hr}$$

$$W_{SO_2}^{13} = W_{SO_2}^{10} = 16.20 \text{ kg/hr}$$

$$W_{NH_3}^{13} = W_{NH_3}^{10} = 8.10 \text{ kg/hr}$$

$$W_I^{13} = W_I^{10} = 399.50 \text{ kg/hr}$$

$$W_{CH_4}^{13} = W_{CH_4}^{10} = 64.80 \text{ kg/hr}$$

$$W_{H_2S}^{13} = W_{H_2S}^{10} = 32.40 \text{ kg/hr}$$

$$W_{Ash}^{13} = 0.00 \text{ kg/hr} \quad (\text{Removed in cyclone})$$

2.9 15th stream weights

Note: Only CO and H₂ are present in Stream 15, with the same values as in Stream 13. All other components are removed in the INY-1 unit.

$$\begin{aligned}W_{CO}^{15} &= W_{CO}^{13} = 10623.66 \text{ kg/hr} \\W_{H_2}^{15} &= W_{H_2}^{13} = 612.04 \text{ kg/hr}\end{aligned}$$

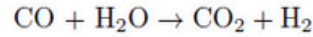
2.10 16th stream weights

Note: Stream 16 consists of the components removed in the INY-1 unit. It carries all the species from Stream 13 except CO and H₂, and includes MDEA from Stream 14.

$$\begin{aligned}W_{SO_2}^{16} &= W_{SO_2}^{13} = 16.20 \text{ kg/hr} \\W_{NH_3}^{16} &= W_{NH_3}^{13} = 8.10 \text{ kg/hr} \\W_{H_2S}^{16} &= W_{H_2S}^{13} = 32.40 \text{ kg/hr} \\W_{CH_4}^{16} &= W_{CH_4}^{13} = 64.80 \text{ kg/hr} \\W_{MDEA}^{16} &= 46778 \text{ kg/hr}\end{aligned}$$

2.11 17th stream weights

Note: Stream 17 consists only of water supplied at 10,000 kg/hr to maximize hydrogen production through the Water–Gas Shift (WGS) reaction:



$$W_{H_2O}^{17} = 10000 \text{ kg/hr}$$

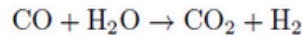
2.12 18th stream weights

Note: In the Water–Gas Shift (WGS) reactor, 90% of the supplied water (from Streams 15 and 17) reacts, while the remaining 10% is unreacted and leaves through Stream 18. Hence, only water is present in Stream 18.

$$\begin{aligned}W_{H_2O}^{18} &= 0.1 \times (W_{H_2O}^{17}) \\&= 0.1 \times (10000) = 1000 \text{ kg/hr}\end{aligned}$$

2.13 20th stream weights

Note: CO from Stream 15 reacts via the Water–Gas Shift (WGS) reaction:



Assumed CO conversion = 90%. Remaining unreacted water has been removed in Stream 18; only the products and leftover CO (and other Stream 15 species) enter Stream 20.

$$\begin{aligned}W_{CO}^{20} &= W_{CO}^{15} - 0.9 W_{CO}^{15} = 0.1 \times 10623.66 = 1062.37 \text{ kg/hr} \\W_{CO_2}^{20} &= W_{CO_2}^{15} + \left(\frac{44}{28}\right) \times (0.9 W_{CO}^{15}) = 0 + \left(\frac{44}{28}\right) \times 9561.29 = 15024.89 \text{ kg/hr} \\W_{H_2}^{20} &= W_{H_2}^{15} + \left(\frac{2}{28}\right) \times (0.9 W_{CO}^{15}) = 612.04 + \left(\frac{2}{28}\right) \times 9561.29 = 1294.99 \text{ kg/hr} \\W_{H_2O}^{20} &= 0.00 \text{ kg/hr} \quad (\text{unreacted water removed in Stream 18})\end{aligned}$$

Other species (passed unchanged from Stream 15):

$$\begin{aligned}W_{SO_2}^{20} &= W_{SO_2}^{15} = 0.00 \text{ kg/hr} \\W_{NH_3}^{20} &= W_{NH_3}^{15} = 0.00 \text{ kg/hr} \\W_I^{20} &= W_I^{15} = 0.00 \text{ kg/hr} \\W_{CH_4}^{20} &= W_{CH_4}^{15} = 0.00 \text{ kg/hr} \\W_{H_2S}^{20} &= W_{H_2S}^{15} = 0.00 \text{ kg/hr} \\W_{Ash}^{20} &= W_{Ash}^{15} = 0.00 \text{ kg/hr}\end{aligned}$$

2.14 21st stream weights

Note: 99.95% of the CO₂ from Stream 20 passes into Stream 21. No H₂ is present in this stream. All other non-zero components from Stream 20 are unchanged and continue to Stream 21.

$$\begin{aligned}W_{CO_2}^{21} &= 0.9995 \times W_{CO_2}^{20} = 0.9995 \times 15024.89 = 15017.38 \text{ kg/hr} \\W_{CO}^{21} &= W_{CO}^{20} = 1062.37 \text{ kg/hr}\end{aligned}$$

2.15 22nd stream weights (Product Stream)

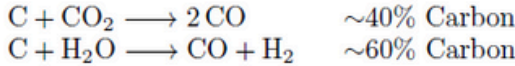
Note: Stream 22 contains the remaining CO₂ not captured in Stream 21, along with all the H₂ produced in Stream 20. This represents the final product stream.

$$\begin{aligned}W_{CO_2}^{22} &= W_{CO_2}^{20} - W_{CO_2}^{21} = 15024.89 - 15017.38 = 7.51 \text{ kg/hr} \\W_{H_2}^{22} &= W_{H_2}^{20} = 1294.99 \text{ kg/hr}\end{aligned}$$

3 Specific Reactions and Calculations

3.1 Gasifier (Single-pass conversion = 100%)

3.1.1 Reactions



3.1.2 Equations

$$\begin{aligned}W_{CO}^{10} &= W_{CO}^7 + ((2)*(0.4)*(W_C^9/12) + (0.6)*(W_{CO}^9/12))*28 \\W_{H_2}^{10} &= W_{H_2}^7 + (0.6)*(W_C^9/12)*2 \\W_{CO_2}^{10} &= W_{CO_2}^9 - (0.4)*(W_C^9/12)*44 \\W_{H_2O}^{10} &= W_{H_2O}^9 - (0.6)*(W_C^9/12)*18 \\W_C^{10} &= 0 \\W_{SO_2}^{10} &= W_{SO_2}^9 \\W_{NH_3}^{10} &= W_{NH_3}^9 \\W_I^{10} &= W_I^9 \\W_{CH_4}^{10} &= W_{CH_4}^9 \\W_{H_2S}^{10} &= W_{H_2S}^9\end{aligned}$$

3.2 WGS Reactor (Single-pass conversion = 90%)

3.2.1 Reaction



$F^{17} = 10000 \text{ kg/hr} \dots$ (Water added through Stream 17)

$$W_{H_2O}^{17} = F^{17}$$

$Z = 0.9 \dots$ (Extent of WGS Reactor)

$$n_{H_2O}^{17} = W_{H_2O}^{17}/18, n_{CO}^{15} = W_{CO}^{15}/28$$

Since, $n_{H_2O}^{17} > n_{CO}^{15}$

Here, CO is the Limiting reactant

So, the reaction proceeds with the full consumption of CO.

$$W_{CO_2}^{20} = (Z) * (n_{CO}^{15}) * (44)$$

$$W_{H_2}^{20} = W_{H_2}^{15} + (Z) * (n_{CO}^{15}) * (2)$$

$$W_{CO}^{20} = W_{CO}^{15} - (Z) * (n_{CO}^{15}) * (28)$$

4 Residual Equations

In this section, we define the final equations that we use to form the required residuals to solve for the unknowns in the matlab code.

Unknowns considered : $F^8, F^9, F^{10}, F^{13}, F^{15}, F^{16}, F^{18}, F^{20}, F^{21}, F^{22}$ (or P)

Equations :

Mass Balance for Stream 8 : (stream 8 contains only CO_2 and C)

$$F^8 = W_{CO_2}^8 + W_C^8 \dots\dots(1)$$

Mass Balance at the Mixer Unit :

$$F^9 = F^6 + F^7 + F^8 \dots\dots(2)$$

Mass Balance at the Gasifier Unit :

$$F^{10} = F^9 \dots\dots(3)$$

Mass Balance at the Cyclone Unit :

$$F^{13} = F^{10} - F^{11} \dots\dots(4)$$

Mass Balance at the INY-1 Unit :

$$F^{16} = F^{13} + F^{14} - F^{15} \dots\dots(5)$$

Mass Balance at the WGS-1 Reactor Unit :

$$F^{20} = F^{17} + F^{15} - F^{18} \dots\dots(6)$$

H_2O Balance for Stream 18 : (stream 18 contains only H_2O)

$$F^{18} = (n_{H_2O}^{17} - Z * (n_{CO}^{15})) * (18) \dots\dots(7)$$

Mass Balance at the FT-1 Unit :

$$F^{21} = F^{20} - P \dots\dots(8)$$

Component-wise mass Balance for Stream 20 :

$$F^{20} = W_{CO}^{20} + W_{CO_2}^{20} + W_{H_2}^{20} \dots\dots(9)$$

Component-wise mass balance for Stream 21 : (stream 21 contains no H_2 ;

all H_2 is in stream 22 or stream P)

$$F^{21} = W_{CO}^{21} + W_{CO_2}^{21} \dots\dots(10)$$

Degree of Freedom Analysis :

Number of unknowns = 10 ($F^8, F^9, F^{10}, F^{13}, F^{15}, F^{16}, F^{18}, F^{20}, F^{21}, F^{22}$ (or P))

Number of equations formed = 10

Degree of Freedom = No. of Unknowns - No. of equations

Therefore, $DOF = 10 - 10 = 0$.

Since, $\text{DOF} = 0$, the system is solvable.

Now, we make 10 residual equations using the 10 equations formed above and find out the value of all the unknown variables using MATLAB.

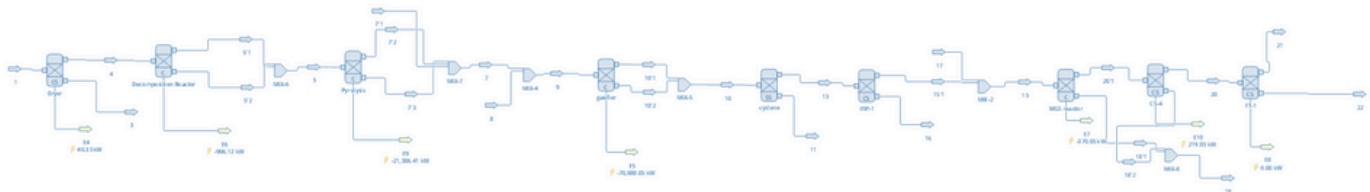
5 Results

The final results from the MATLAB code come out to be :

$$\begin{aligned} F^8 &= 3257 \text{ kg/hr} \\ F^9 &= 11757 \text{ kg/hr} \\ F^{10} &= 11757 \text{ kg/hr} \\ F^{13} &= 11357 \text{ kg/hr} \\ F^{15} &= 11236 \text{ kg/hr} \\ F^{16} &= 46900 \text{ kg/hr} \\ F^{18} &= 3853 \text{ kg/hr} \\ F^{20} &= 17382 \text{ kg/hr} \\ F^{21} &= 16080 \text{ kg/hr} \\ F^{22}(\text{or } P) &= 1302 \text{ kg/hr} \end{aligned}$$

1302 kg/hr of Hydrogen Gas is produced.

DWSIM :



PROPERTIES TABLE			
1	Mass Flow	7002.63	kg/h
10	Mass Flow	11359.7	kg/h
3	Mass Flow	1500	kg/h
4	Mass Flow	5502.63	kg/h
5	Mass Flow	5502.63	kg/h
7	Mass Flow	8100.5	kg/h
8	Mass Flow	3259.18	kg/h
9	Mass Flow	11359.7	kg/h
11	Mass Flow	0	kg/h
13	Mass Flow	11359.7	kg/h
15	Mass Flow	11223.5	kg/h
16	Mass Flow	136.159	kg/h
17	Mass Flow	10000	kg/h
18	Mass Flow	3914.52	kg/h
20	Mass Flow	17309	kg/h
21	Mass Flow	16007.7	kg/h
22	Mass Flow	1301.27	kg/h

Differences in Simulation and the MATLAB

1. In DWSIM, mass flow rate of Ash(inorganic matter), in all the relevant streams (1,4,6,9,10,11) and Methyldiethanolamine in 14 and 16 streams are 0 kg/h respectively. Hence streams 1,4,9,10,16 have lesser flowrates and there is no 6th and 12th stream and 16th stream has 46778 kg/h less flow rate and 11th stream has 0 mass flow rate.

only property package used is Rault's law

Conclusion

- 1.This project successfully designed and modeled a complete, multi-stage process for producing high-purity hydrogen from a biomass feedstock.
- 2.This analysis confirmed the model was solvable with a Degree of Freedom (DOF) of zero, validating its theoretical feasibility.
- 3.Based on the MATLAB simulation, the process is capable of producing a final product stream at a rate of 1302 kg/hr.
4. The final stream is high-purity, consisting of 99.9% of H₂ and only 0.01% of residual CO₂.